HW 6

3)

a) $RSS = (y_1 - (\beta_1 x_{11} + \beta_2 x_{12}))^2 + (y_2 - (\beta_1 x_{21} + \beta_2 x_{22}))^2$

$$\text{Minimize}_{\beta_1 \beta_2} \{(y_1 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_1 x_{21} - \beta_2 x_{22})^2 + \lambda(\beta_1^2 + \beta_2^2)\}$$

since $x_{11} = x_{12}$ and $x_{21} = x_{22}$

I can shorten to x_{11} & x_{12} to x_1 and x_{21} & x_{22} to x_2

$$\text{Minimize}_{\beta_1, \beta_2} \{(y_1 - \beta_1 x_1 - \beta_2 x_1)^2 + (y_2 - \beta_1 x_2 - \beta_2 x_2)^2 + \lambda(\beta_1^2 + \beta_2^2)\}$$

$$\Rightarrow \text{Minimize}_{\beta_1, \beta_2} \{(y_1 - (\beta_1 + \beta_2) x_1)^2 + (y_2 - (\beta_1 + \beta_2) x_2)^2 + \lambda(\beta_1^2 + \beta_2^2)\}$$

b) The residual terms only depend on $\beta_1 + \beta_2$. Swapping the values of β_1 and β_2 does not change anything.

$$\beta_1 + \beta_2 = \beta_2 + \beta_1 \text{ and } (\beta_1^2 + \beta_2^2) = (\beta_2^2 + \beta_1^2)$$

L2 penalty creates a convex optimization with a unique global min. Swapping β_1 and β_2 won't change anything

c) Minimize $_{\beta_1, \beta_2} \{(y_1 - \beta_1 x_{11} - \beta_2 x_{12})^2 + (y_2 - \beta_1 x_{21} - \beta_2 x_{22})^2 + \lambda(|\beta_1| + |\beta_2|)\}$

$\Rightarrow$

Minimize $_{\beta_1, \beta_2} \{(y_1 - (\beta_1 + \beta_2)x_1)^2 + (y_2 - (\beta_1 + \beta_2)x_2)^2 + \lambda(|\beta_1| + |\beta_2|)\}$

d) Since the two variable are identical, the model only cares about their combined sum $\beta_1 + \beta_2$. The absolute value penalty treats any split of that sum equally as long as they have the

same sign. Because of this, there are infinite possible combinations that give the exact same best score, so the coefficients are not unique

Like the model only cares about $\beta_1 + \beta_2 = 3$ (for example)

the penalty is $|\beta_1| + |\beta_2|$

you could do $\beta_1 = 3$ $\beta = 0$ or $\beta_1 = \beta_2 = 1.5$

or $\beta_1 = 0$ $\beta_2 = 3$. This is not a unique solution